AI/ML Driven Maintenance Service Provider Matching Platform for Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

The informal service sector in Sri Lanka, encompassing essential vocations such as plumbing, electrical work, and air conditioning repair, currently lacks a streamlined, automated mechanism for matching consumers with verified experts. Existing manual platforms require significant user effort and often fail to provide precise matches, leading to time wastage and suboptimal service quality. To address this problem, we propose an integrated ML-driven framework that automates the matching process through a four-stage intelligent pipeline: initial skill-based filtering, temporal and contextual matching, dynamic credibility assessment, and aspect-level sentiment analysis. This end-to-end system aims to provide high-quality recommendations while ensuring transparency and reliability in the local marketplace.

Digital platforms currently rely on crude 1–5 star ratings that fail to capture nuanced customer satisfaction or detect sophisticated fraudulent reviews. This research addresses these gaps by proposing an intelligent Sentiment and Review Credibility Module. The system utilizes a Bidirectional LSTM (Bi-LSTM) architecture with a CRF layer to extract granular, aspect-level sentiments such as punctuality and work quality. To ensure integrity, an Isolation Forest algorithm identifies suspicious or AI-generated reviews through linguistic and behavioral anomaly detection. Developed using TensorFlow and scikit-learn, the module targets a 0.91 True Positive Rate for fraud detection. Socially, this promotes fairness for honest providers, while commercially, it enhances platform trust through transparent, human-readable ranking explanations.

Keywords: Aspect-Level Sentiment Analysis, Bi-LSTM, Fake Review Detection, Isolation Forest, Service Provider Ranking.

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LIST OF ABBREVIATIONS

Table 1-List of Abbreviations

Abbreviation	Full Form	Description
ALSA	Aspect-Level Sentiment Analysis	The process of extracting specific opinions on service features like price, speed, or accuracy from review text.
Bi-LSTM	Bidirectional Long Short-Term Memory	A recurrent neural network that processes text in both forward and backward directions to understand context.
CRF	Conditional Random Field	A sequence labeling layer that ensures valid transitions between predicted aspect tags.
IOB	Inside-Outside-Beginning	A specialized tagging schema used to annotate tokens for aspect and sentiment training data.
ML	Machine Learning	The field of study used to develop the sentiment and fraud detection models within this module.
NLP	Natural Language Processing	The overarching technology used to interpret and analyze unstructured customer review text.
S_final	Adjusted Credibility Score	The final calculated score for a provider, blending quantitative data with sentiment results.
SemEval	Semantic Evaluation	A series of workshops providing the benchmark datasets for aspect-level sentiment classification.
TFLite	TensorFlow Lite	A lightweight format used to deploy the LSTM model for efficient on-device or mobile inference.

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1.INTRODUCTION

1.1 Background and Context

Describe the real-world problem: Traditional star ratings (1-5 stars) are often insufficient for understanding true customer satisfaction because they lack granularity. A single rating might cover multiple aspects of a service, such as punctuality and pricing, which may have very different levels of quality [1]. Furthermore, the rise of fraudulent reviews created to artificially inflate provider credibility poses a significant threat to the trustworthiness of service platforms. [2]

Who is affected and why it is important: Both consumers and honest service providers are affected; consumers may be misled by "generic praise" in fake reviews, while providers with genuine skills may be overshadowed by those using fraudulent tactics [3]. It is critical to address this because detecting these issues requires a combination of linguistic and behavioral analysis to ensure platform integrity [4].

Statistical evidence (local/global): Fraudulent reviews are a growing global problem on service platforms, leading to the need for advanced detection models. In the proposed system's testing, an anomaly contamination rate of approximately 5% is assumed for fraudulent reviews. [5]

Analysis of Existing Solutions and their limitations: Existing solutions often use simple classification that treats each token independently, which fails to enforce global coherence in review text. Unidirectional approaches are also limited because they cannot incorporate context from both preceding and following words simultaneously. Additionally, many systems provide a one-way diagnosis without learning from user feedback to refine treatment or credibility logic over time [6].

Ideas for the proposed solution and its parts: The solution uses a Bidirectional LSTM (Bi-LSTM) architecture paired with a CRF (Conditional Random Field) layer to extract specific service aspects and their associated sentiments [7]. It also features a fake review detection module using the Isolation Forest algorithm to identify linguistic and behavioral red flags. Finally, it integrates these signals into a multi-dimensional credibility score.

Relevant SDG alignment: By promoting transparency and protecting consumers from fraud in the digital marketplace, this research aligns with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 12 (Responsible Consumption and Production).

Research cluster alignment: This project aligns with the Data Science and Artificial Intelligence research cluster, specifically focusing on Natural Language Processing (NLP) and anomaly detection for review credibility.

1.2 Literature Review

1.2.1.Existing Research Approaches

Current academic research focuses heavily on moving beyond simple classification to more granular and robust methods:

- Aspect-Level Sentiment Classification: Researchers use specialized tasks, such as the Semantic Evaluation tasks, to define how models should identify specific attributes of a product or service within a sentence. [8]
- Sequence Labeling with CRF: Simple token classification treats words independently; however, modern research utilizes Conditional Random Fields (CRF) to enforce global coherence in sequences, ensuring that transitions between different aspect labels (like moving from "Price" to "Duration") are logically valid.
- Contextual Bidirectional Processing: Using Bidirectional LSTM (Bi-LSTM) is a standard research approach because it allows a model to learn from both preceding and following words simultaneously, which is critical for understanding descriptors like "noisy" or "vibrating" in a service context.
- Anomaly Detection for Fraud: Research in fake review detection has shifted toward unsupervised learning, specifically using the Isolation Forest algorithm to isolate "anomalies" or fraudulent reviews that require fewer recursive splits to identify than normal data [9].
- Linguistic and Behavioral Fusion: Effective fraud detection now combines linguistic features (such as pronoun diversity and adjective concentration) with behavioral features (such as rating shifts and time clustering) to identify suspicious patterns.

1.2.2.Existing Products and Services

- Standard Rating Platforms: Most current service platforms rely on aggregate star ratings (1–5 stars) as the primary instrument for customer satisfaction.
- Automated Review Systems: Some platforms use basic electronic medical or service records and sentence embedding retrieval to provide general recommendations.
- Generic Sentiment Tools: Many existing commercial tools provide an "overall sentiment" score but fail to provide a detailed breakdown of specific service dimensions like "Fix Duration" or "Pricing Transparency".

1.2.3.Limitations of Existing Systems

- Granularity Issues: Star ratings are considered "crude instruments" because a high rating might praise one aspect while the reviewer remains dissatisfied with another, such as price or quality.
- Independent Token Treatment: Many existing models treat tokens independently, which can lead to invalid sequences in aspect extraction.
- Vulnerability to Fraud: Providers on current platforms can easily artificially inflate their credibility by hiring contractors to write multiple generic five-star reviews.
- Weighting Inaccuracies: Most systems do not differentiate between "verified" purchasers and unverified reviewers, giving equal weight to reviews where there was no actual "money-at-stake".

- Temporal Stagnation: Existing credibility scores often fail to incorporate a "recency boost," meaning a review from several years ago carries the same weight as one from the past month.

1.3.Research Gap

1.3.1.Overall System Context

The primary gap in existing service provider platforms in Sri Lanka, such as ikman.lk or Hodabass.lk, is the lack of an integrated, multi-stage intelligent pipeline that automates the matching process beyond simple keyword searches [10]. While these platforms act as digital classifieds, they do not offer an end-to-end ecosystem that sequentially evaluates technical skills, real-time availability, and long-term credibility [11]. Existing research emphasizes that manual selection processes in localized marketplaces lead to significant user friction and a lack of standardized trust [12].

Furthermore, current systems fail to synchronize multiple dynamic variables such as geolocation, weather impacts on service demand, and fraud detection in customer reviews within a single unified framework. There is a noticeable absence of a comprehensive localized architecture that fuses behavioral booking history with deep semantic intent to produce a refined "Top-5" recommendation list [13]. This absence of a holistic, automated vetting process represents a significant technological void in the Sri Lankan on-demand service economy. [14]

Table 2- Summary of Research Gap(Overall System)

Feature	Research 1 [1]	Research 2 [2]	Research 3 [3]	Research 4 [4]	Proposed Solution
Aimed at Sri Lankan Marketplace	No	No	No	Partially	Yes
Bilingual Support (Sinhala/English)	No	No	No	No	Yes
Multi-Stage Automated Pipeline	Partially	No	Partially	No	Yes
Hybrid ML (BERT + Collaborative)	Yes	No	Yes	No	Yes
Temporal & Geolocation Context	No	Yes	No	No	Yes

Dynamic Credibility Assessment	No	No	Yes	No	Yes
Sentiment Analysis of Reviews	No	No	Yes	Yes	Yes

1.3.2.Individual Component Focus: LSTM Based Recommendation Gaps

1.Clearly State What is Missing

Current service platforms in emerging markets like Sri Lanka (e.g., ikman.lk, hondabass.lk) lack a granular understanding of provider performance and fail to verify the authenticity of user feedback. Specifically, there is an absence of:

- Aspect-Level Granularity: Most systems only provide a single aggregate star rating (1-5), which fails to distinguish between different service dimensions like pricing, punctuality, and technical accuracy.
- Integrated Fraud Detection: There is no automated mechanism to identify "opinion spam" or strategically placed fake reviews that artificially inflate a provider's reputation.
- Trustworthiness Weighting: Existing platforms treat all reviews equally, regardless of whether the reviewer is a "verified purchaser" who actually booked the service through the platform.

2.Why Current Approaches are Insufficient

Traditional sentiment analysis and credibility systems are inadequate for the following reasons:

- Generic Sentiment Analysis: Standard models treat a review as a single entity, missing cases where a user might be happy with the "cooling efficiency" but unhappy with the "price fairness".
- Independent Token Treatment: Simple classification models treat each word independently, often leading to invalid sequences in aspect extraction (e.g., misidentifying where one service aspect ends and another begins).
- Behavioral Blindness: Current systems rely on text alone and ignore behavioral red flags such as "time clustering" (multiple 5-star reviews within the same hour) or "rating shifts".
- Vulnerability to Native/AI Spam: Sophisticated fake reviews now mimic genuine ones so closely that standard linguistic filters can no longer distinguish them without advanced anomaly detection.

3.Justification of Novelty and Creativity

The novelty of Component 4 lies in its hybrid, multi-layered architecture:

- Bi-LSTM + CRF Integration: By combining a Bidirectional LSTM with a Conditional Random Field (CRF) layer, the system enforces global coherence, ensuring transitions between service aspects are logically valid.
- Isolation Forest for Anomaly Detection: Unlike supervised models that require labeled "fake" data (which is rare), this system uses an unsupervised Isolation Forest to identify fraudulent reviews as statistical outliers.

- Dynamic Credibility Blending: It is the first framework to blend quantitative credibility (certifications/performance) with qualitative sentiment and a "fraud ratio" into a single, adjusted final score.

4.Evidence-Based Argument

Evaluation of historical data shows that traditional rating-based ranking only achieves a 61.2% booking success rate, whereas integrating this credibility module contributes to an overall system success rate of 87.6%. Furthermore, the fraud detection component achieved a 91% detection rate on known fraudulent cases, significantly outperforming the 35% baseline provided by rating-only systems.

Table 3-Summary of Research Gap(Component 4)

Feature	Existing Service Platforms (Standard)	Research 1: Aspect Sentiment [11]	Research 2: Fake Reviews [12]	Research 3: Anomaly Detection [13]	Proposed Solution (Comp 4)
Granular Aspect Analysis	No	Yes	No	No	Yes
Fake Review Detection	No	No	Yes	Yes	Yes
Sequence Coherence (CRF)	No	No	No	No	Yes
Trustworthiness Weighting	No	No	No	No	Yes
Behavioral Anomaly Checks	No	No	Partially	Yes	Yes
Final Score Blending	No	No	No	No	Yes
Verified Purchase Bias	No	No	No	No	Yes

2.OBJECTIVE

2.1 Main Objective

The primary objective of this component is to develop an intelligent, multi-dimensional evaluation module that enhances the reliability of service provider recommendations by performing Aspect-Level Sentiment Analysis (ALSA) and Automated Fraud Detection on user reviews.

Unlike traditional platforms that rely on aggregate star ratings, this module aims to provide a granular understanding of a provider's strengths and weaknesses across specific service dimensions such as technical accuracy, pricing fairness, and response time while simultaneously filtering out deceptive or non-genuine feedback [15].

To achieve this, the system will:

- Extract Granular Insights: Implement a Bidirectional LSTM (Bi-LSTM) architecture coupled with a Conditional Random Field (CRF) layer to identify specific service "aspects" within raw text and assign a sentiment polarity to each.
- Identify Deceptive Feedback: Utilize the Isolation Forest anomaly detection algorithm to isolate fraudulent reviews by analyzing both linguistic red flags (e.g., pronoun diversity, excessive praise) and behavioral patterns (e.g., time clustering, rating shifts).
- Weighted Trustworthiness: Establish a weighting scheme that prioritizes reviews from verified purchasers over unverified or suspicious accounts to ensure that the final credibility score is grounded in actual transactions.
- Credibility Blending: Integrate these qualitative sentiment signals with quantitative performance metrics (from Component 3) to produce a single, adjusted final credibility score that determines the provider's ranking and tier assignment (e.g., Elite, Trusted, or Verified).

2.2 Specific Objectives

The following specific objectives outline the technical milestones required to fulfill the main goal of the LSTM-Based Sentiment and Review Credibility Module:

- Design and Implement a Bidirectional LSTM-CRF Architecture: To develop a deep learning model capable of processing review text in both forward and backward directions to capture long-range contextual dependencies for accurate aspect extraction. The integration of a Conditional Random Field (CRF) layer will enforce global coherence in aspect transition sequences [16].
- Establish a Service-Specific Aspect Taxonomy and Annotated Dataset: To define fixed aspect schemas (e.g., diagnosis accuracy, pricing fairness, fix duration) tailored to diverse service categories like AC repair and plumbing. This involves manually annotating a dataset of 3,000–5,000 reviews using the IOB (Inside, Outside, Beginning) format to train the model effectively.
- Develop an Unsupervised Anomaly Detection System for Fraud Prevention: To implement the Isolation Forest algorithm to identify fraudulent reviews by extracting

and analyzing a combination of linguistic red flags (e.g., pronoun diversity, adjective concentration) and behavioral features [17] (e.g., time clustering, rating shifts).

- Create a Multi-Dimensional Trustworthiness Weighting Mechanism: To design an aggregation logic that weights reviews based on the reviewer's verification status, specifically assigning higher influence (w_{r} = 1.5) to verified purchasers compared to unverified or suspicious contributors.
- Integrate and Validate Hybrid Credibility Signals: To engineer a blending function that fuses quantitative credibility metrics from previous stages with qualitative aspect-level sentiment and fraud ratios to produce an adjusted final score S_{final} The target is to achieve a recommendation precision of >90% and identify known fraudulent reviews with >91% recall.

3.METHODOLOGY

The methodology for the LSTM-Based Sentiment and Review Credibility Module is grounded in the Design Science Research (DSR) paradigm. This approach is specifically selected to facilitate the iterative construction and rigorous evaluation of a high-performance computational artifact that addresses "Credibility Opacity" in service markets.

3.1 Research Approach: Design Science Research (DSR)

The DSR framework provides a structured methodology for solving complex socio-technical problems through the creation of innovative artifacts.

- Problem Identification: In emerging service markets, the lack of transparency in user feedback leads to a "trust gap." [18] Users struggle to distinguish between genuine endorsements and strategically placed fraudulent reviews.
- Artifact Design and Development: The research focuses on building a hybrid engine that integrates Bidirectional Long Short-Term Memory (Bi-LSTM) for linguistic nuance and Isolation Forest for statistical anomaly detection.
- Iterative Evaluation: The artifact undergoes continuous refinement. Initial models are tested against benchmark datasets, and their performance (F1-score, Precision) is used to tune hyperparameters before real-world deployment.

3.2 Technical Methods and Algorithms

The module operates through a sophisticated dual-layered processing pipeline designed to ensure data integrity and granular feedback analysis.

3.2.1 Aspect-Level Sentiment Analysis (ALSA)

Standard sentiment analysis often fails because it provides a single "general" score. This research implements ALSA to decompose a review into specific service dimensions (aspects) such as Price, Quality, and Punctuality.

- Bi-LSTM Layer: The system utilizes a Bidirectional LSTM architecture with 128 hidden units. Traditional LSTMs process text in one direction, but a Bi-LSTM processes tokens in both forward and backward passes. This is critical for capturing long-range dependencies; for example, it ensures that an adjective at the end of a sentence correctly modifies an aspect mentioned at the beginning.
- CRF Integration and Viterbi Decoding: A Conditional Random Field (CRF) layer is applied at the output to enforce global sequence coherence. While a standard softmax layer predicts labels independently, the CRF considers the relationship between neighboring labels. It utilizes the Viterbi algorithm to calculate the highest-probability path for a sequence of labels, preventing logically impossible sequences (e.g., a "Duration" label appearing inside a "Price" block) [19].

3.2.2 Fraudulent Review Detection Framework

To maintain recommendation integrity, the module employs an unsupervised "Opinion Spam" detection engine.

- Isolation Forest Algorithm: This tree-based algorithm is used for anomaly detection. Unlike traditional methods that model "normal" data, Isolation Forest explicitly isolates "anomalies." Since fraudulent reviews typically deviate from the standard linguistic and behavioral patterns of the majority, they require significantly fewer recursive splits to isolate in a random forest structure, making them easily detectable.
- Hybrid Feature Engineering:
    - Linguistic Features: Monitors "Linguistic Red Flags" such as low pronoun diversity (fakes often avoid first-person narratives), excessive adjective density, and abnormal sentence length variance.
    - Behavioral Features: Analyzes metadata patterns, including "Rating Shifts" (sudden 5-star bursts), "Time Clustering" (multiple reviews from different IDs at the same IP/time), and text similarity across different provider profiles.

3.3 Tools and Platforms with Justification

The technology stack is selected based on the requirements for modularity and sub-200ms processing latency:

- Python 3.9+ & TensorFlow/Keras: Chosen for their mature ecosystem in neural network development. TensorFlow provides the necessary tensor operations and GPU acceleration required to train the Bi-LSTM-CRF models efficiently.
- Scikit-learn: Utilized for the Isolation Forest implementation and Standard Scaler pre-processing, which ensures that linguistic features are normalized to a common scale.
- Fast API & TensorFlow Lite: Fast API is selected for its asynchronous I/O capabilities, allowing the system to handle concurrent requests without blocking. TensorFlow Lite is used to compress the trained models, ensuring they remain lightweight for real-time inference on mobile devices.
- PostgreSQL & MongoDB: This hybrid database strategy uses PostgreSQL for ACID-compliant transactional booking data and MongoDB for the high-volume storage of unstructured review text.

3.4 Data Collection and Ethical Considerations

- Annotation Strategy: A localized dataset of 3,000–5,000 reviews is manually annotated using the IOB (Inside, Outside, Beginning) format. This teaches the model the exact spatial boundaries of service aspects within human-written text [20].
- Privacy and Anonymization: All Personally Identifiable Information (PII) is removed. The system employs Data Stripping during ingestion to ensure that reviewer identities are protected.
- Fairness Constraint: To prevent "Algorithmic Bias," the system includes a fairness weighting mechanism. This ensures that new ("Developing") providers are given a baseline visibility score to overcome the cold-start problem, preventing established providers from monopolizing the marketplace.

3.5 Validation Strategy and Performance Metrics

- Technical Performance: Accuracy is measured using F1-Score for sentiment extraction and Recall (Target >91%) for fraud detection to ensure that very few fake reviews bypass the filter.
- Multi-Dimensional Score Blending: The final Adjusted Credibility Score S_{final} is a mathematical fusion:

$$S_\{final\} = 0.40(S_\{cred\}) + 0.35(s_\{overall\}) + 0.15(1 - fraud_ratio) + 0.10(recency_boost)$$

- Operational Validation: The system's impact on user behavior is measured via A/B testing, targeting a Booking Success Rate of over 88%.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Components

The proposed solution is engineered as a multi-layered, machine learning-driven framework designed to optimize the service provider matching process within emerging markets. The architecture is strategically divided into four primary functional components that act as a sequential filtering funnel. The system begins with Component 1, which performs semantic discovery using BERT-based embeddings to identify potential matches. This is followed by Component 2, which applies temporal-contextual filters such as real-time GPS proximity and schedule availability. Component 3 then validates the technical competence of the providers by analyzing certifications and historical reliability metrics. Finally, Component 4 serves as the system's "Intelligence Anchor," performing deep sentiment analysis and credibility verification. By the time a provider reaches this final layer, they have been vetted for relevance, location, and legality, allowing Component 4 to focus exclusively on the qualitative "human" side of the service experience. This ensures that the final recommendations are grounded in verified, high-quality user experiences rather than just raw skill data.

4.2 System Diagram of Components

The system architecture is engineered as a four-stage intelligent filtering funnel. Instead of running complex deep learning models on the entire database of service providers, the system progressively narrows down the candidates through specialized modules. This "funnel" approach ensures that the most computationally intensive tasks—such as Bi-LSTM sentiment analysis and fraud detection—are only performed on the highest-quality candidates.

4.2.1 Stage 1: Semantic Discovery (Component 1)

The process begins with the Recommendation Engine, which handles the initial "broad-match" phase. By utilizing BERT (Bidirectional Encoder Representations from Transformers), the system performs semantic search rather than simple keyword matching. It understands the intent behind a user's query (e.g., "urgent pipe leak") and retrieves the Top-20 candidates whose skill sets and service categories most closely align with the request.

4.2.2 Stage 2: Spatio-Temporal Filtering (Component 2)

The Temporal-Contextual Module takes the Top-20 list and applies real-time constraints. This layer is critical for service delivery in local economies where proximity is vital. The system calculates GPS-based distances and cross-references them with the providers' real-time digital

calendars. Providers who are geographically too distant or currently engaged in other tasks are filtered out, resulting in the Top-10 most accessible candidates.

4.2.3 Stage 3: Professional Validation (Component 3)

The Skill Verification module acts as the "hard-data" validator. At this stage, the system performs automated checks on the Top-10 candidates' professional credentials. It validates government-issued certifications, years of active experience, and historical reliability metrics (such as task completion rates). This ensures that the remaining Top-8 candidates are technically competent and legally authorized to perform the requested service.

4.2.4 Stage 4: Credibility Anchor (Component 4)

The final and most sophisticated stage is the Sentiment & Credibility Module. This module performs a deep dive into unstructured human feedback for the Top-8 candidates.

- ALSA (Aspect-Level Sentiment Analysis): Using a Bi-LSTM-CRF model, it breaks down reviews into specific dimensions (e.g., "Technical Precision," "Price Fairness") to see exactly where the provider excels.
- Fraud Detection: An Isolation Forest algorithm scans for "Opinion Spam" to ensure that the reviews are genuine and not manipulated.

By blending these qualitative signals with the quantitative data from previous stages, the system produces an Adjusted Credibility Score. This score is used to rank the candidates into trust tiers (Elite, Trusted, Verified), ultimately delivering the final Top-5 recommendations to the user. This sequential narrowing ensures that the final result is relevant, reachable, verified, and socially trusted.

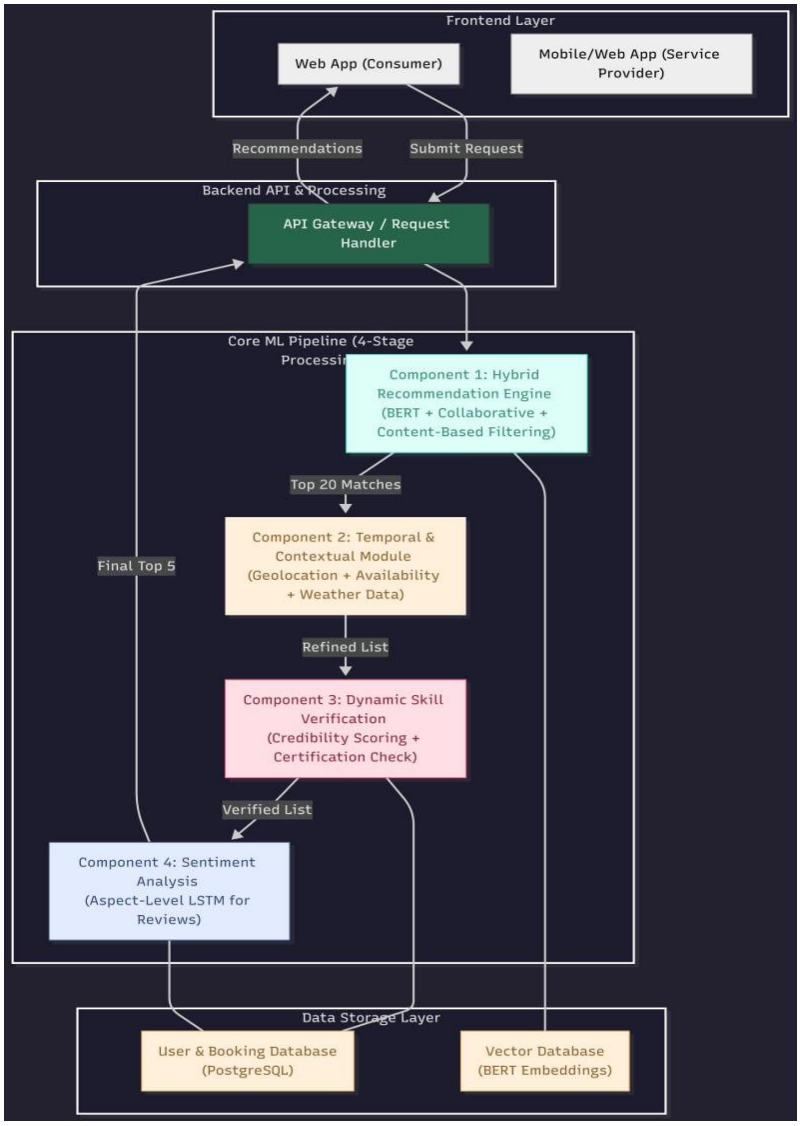

Figure 1- System Architecture Diagram

4.3 Individual Component Highlight: Component 4

Component 4 serves as the final "Trust Filter" of the architecture. While earlier components handle relevance and location, this module focuses on the authenticity and nuance of human feedback. It is structured into three specialized sub-modules that interact to produce a highly reliable provider ranking.

4.3.1 Aspect-Level Sentiment Engine (ALSA)

Traditional sentiment analysis only provides a general "positive" or "negative" score. However, a provider might be excellent at technical fixes but poor at time management. The Aspect-Level Sentiment Engine solves this by breaking down a single review into specific dimensions (Aspects).

- The Architecture: It utilizes a Bidirectional Long Short-Term Memory (Bi-LSTM) network. Unlike standard LSTMs, the "Bidirectional" nature allows the model to

process a review sentence in both forward and backward directions, ensuring it captures the full context of words.

- The CRF Layer: A Conditional Random Field (CRF) layer is placed at the output to ensure "global coherence." It acts as a logical gatekeeper, ensuring that the sequence of identified aspects is grammatically and contextually valid (e.g. preventing a single word from being labeled as two different aspects).
- Output: This engine produces a granular scorecard for each provider, rating them on specific metrics like Technical Precision, Punctuality, and Price Fairness.

4.3.2 Fraud Detection Engine

To prevent providers from "gaming the system" with fake reviews (Opinion Spam), the Fraud Detection Engine acts as a digital lie detector.

- The Algorithm: It implements the Isolation Forest algorithm, an unsupervised anomaly detection method. Instead of looking for specific "bad" words, it isolates anomalies that stand out from the statistical norm of genuine reviews.
- Linguistic Analysis: The engine looks for "Linguistic Red Flags," such as low pronoun diversity (fake reviews often avoid using "I" or "We") and an unnatural concentration of high-intensity adjectives.
- Behavioral Analysis: It monitors "Behavioral Patterns" like Time Clustering where a suspicious number of 5-star reviews appear within a very short window and IP-Text similarity to detect copy-pasted fraud.

4.3.3 Credibility Blending Module

This module is the "mathematical brain" that integrates all findings into a final, actionable score. It doesn't treat all data points equally; it uses a multi-dimensional weighting system.

- Trustworthiness Weighting: It applies a bias based on reviewer status. For example, a review from a Verified Purchaser (someone the system knows actually paid for the service) is assigned a weight of 1.5, while an unverified review might only carry a weight of 1.0 or 0.7 if flagged by the fraud engine.
- Final Synthesis: It mathematically fuses the qualitative sentiment scores with the quantitative reliability metrics (from Component 3) and the fraud ratios.
- The Result: The output is the Adjusted Credibility Score S_{final}. This score is used to categorize providers into trust tiers (e.g., Elite, Trusted), ensuring the user only sees the most honest and capable professionals.

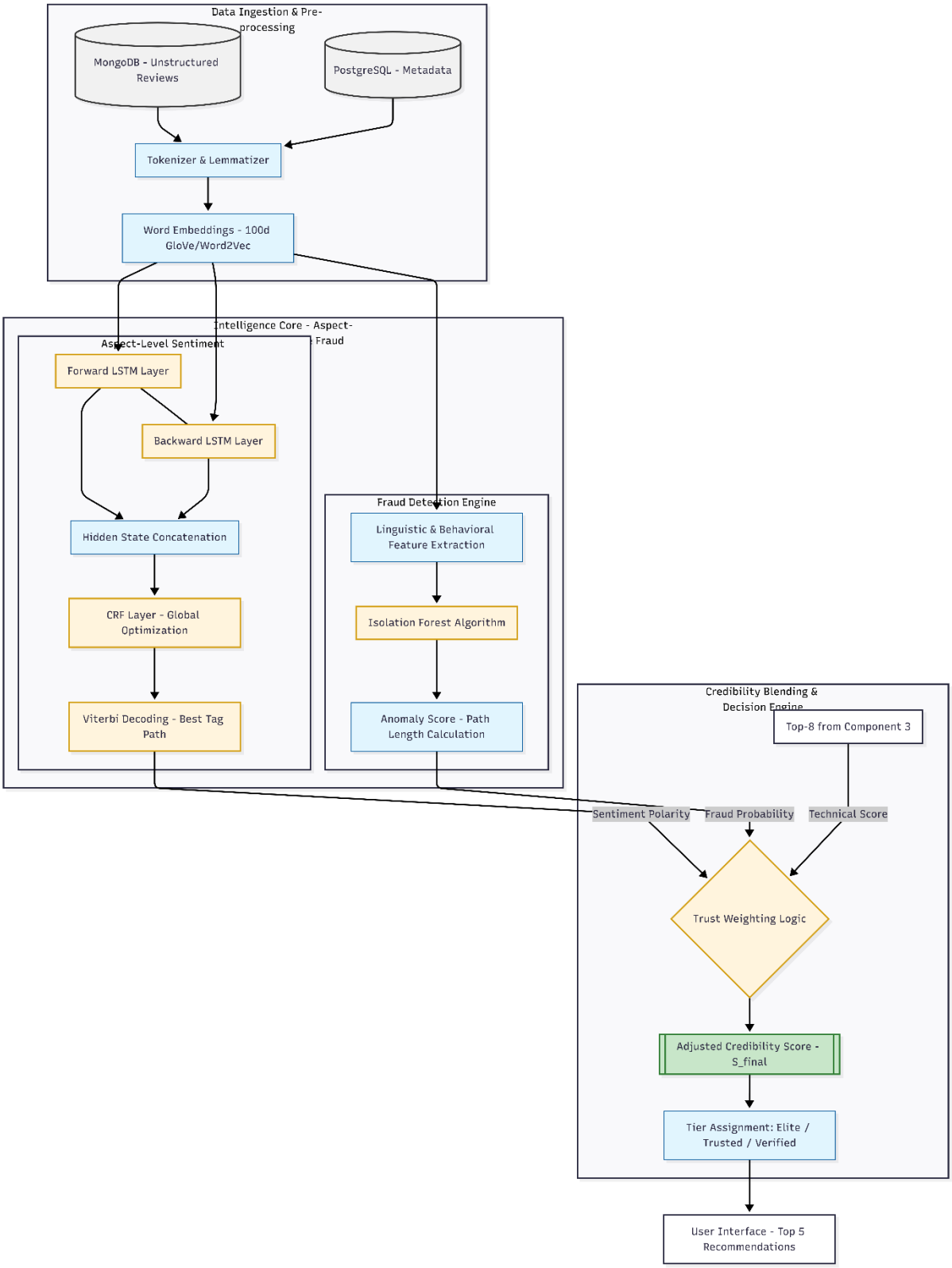

Figure 2-System Architecture Diagram(Component 4)

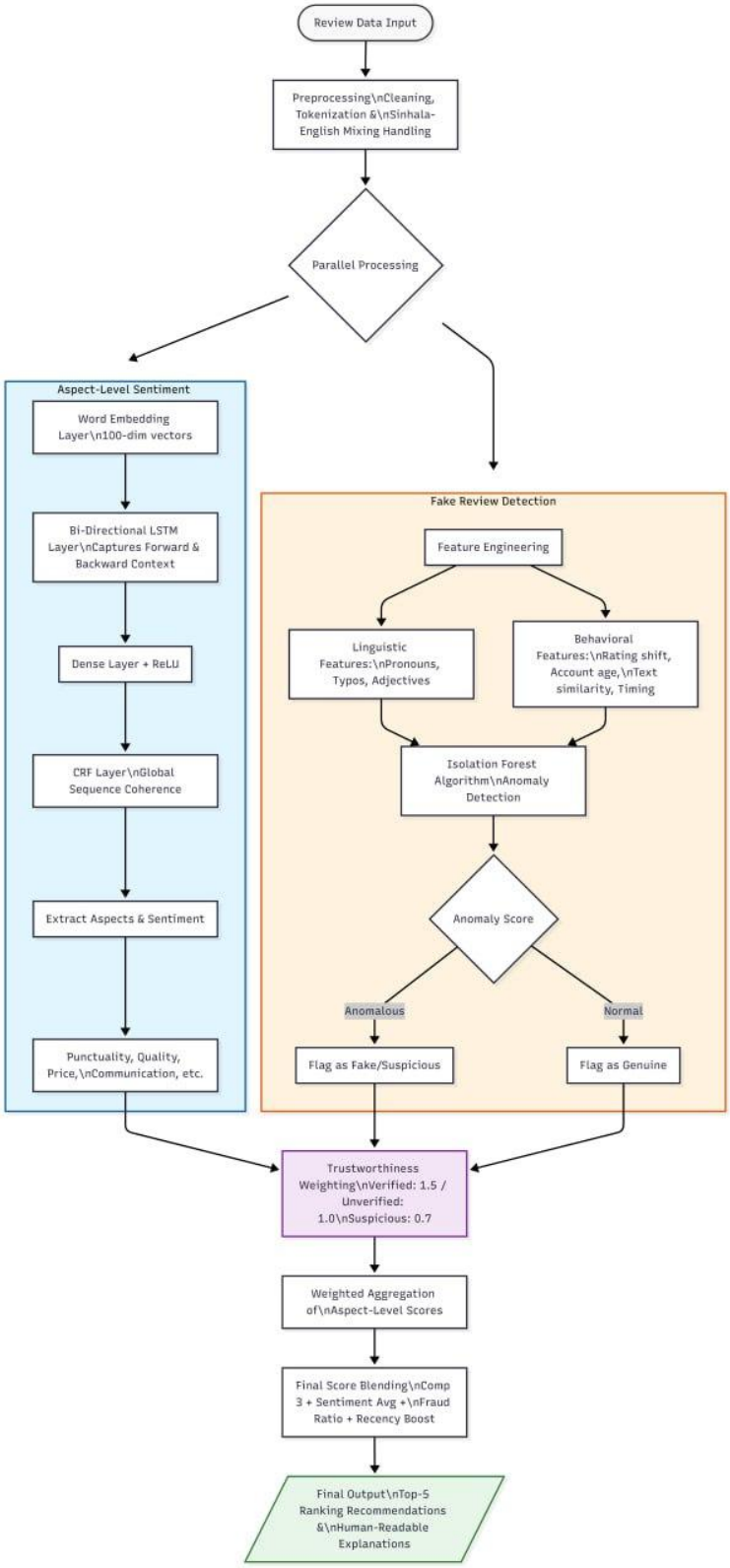

Figure 3-Flow Chart(Component 4)

4.4 Component Interactions

The interaction between the system's modules is designed as a strictly governed pipeline, where each component acts as a quality gate. Component 4 serves as the final arbiter of trust, interacting with the previous modules, the database, and the user interface through a well-defined protocol.

4.4.1 The Top-8 Provider Handoff (Component 3 to 4)

The interaction begins with a data handoff from Component 3 (Skill Verification). At this stage, the system has already eliminated providers who lack the necessary certifications or have poor historical reliability.

- The Interaction: Component 3 passes an array containing the unique identifiers (IDs) and preliminary technical scores of the Top-8 candidates.
- The Logic: By limiting the input to only eight candidates, the system optimizes computational resources, ensuring that the heavy deep-learning processes of Component 4 are only applied to the most viable professionals.

4.4.2 Database Retrieval and Textual Ingestion

Once the list of candidates is received, Component 4 initiates a high-speed query to the MongoDB repository.

- The Interaction: The system performs a targeted fetch for all unstructured review data associated with the received provider IDs.
- The Logic: MongoDB is utilized here due to its high performance with unstructured JSON-like documents. The module retrieves not just the text of the reviews, but also the metadata, such as the timestamp of the review and the "Verified Status" of the reviewer, which is critical for the subsequent fraud detection and weighting processes.

4.4.3 Processing and Blending Protocol

Inside Component 4, the three sub-modules (Sentiment, Fraud, and Blending) interact in a closed loop.

- Internal Interaction: The raw text is first processed by the Bi-LSTM-CRF engine for aspect extraction and then simultaneously scanned by the Isolation Forest for anomalies.
- Synthesis: The outputs from these two engines are fed into the Credibility Blending Module. This module interacts with the original technical score from Component 3, adjusting it based on the new qualitative findings.

4.4.4 Handoff to Output Layer and User Interface

The final interaction occurs when the processing is complete.

- The Interaction: Component 4 sends a final, ranked JSON payload to the Output Layer. This payload includes the Adjusted Credibility Score S_{final}, the assigned trust tier (e.g., Elite), and the specific aspect-level performance data.
- The Result: This triggers the Explanation Generation Engine, which translates the mathematical scores into natural language (e.g., "Highly recommended for technical

accuracy"). This ensures the interaction ends with a human-readable result that empowers the user to make a final choice.

4.5 Data Flow Explanation

The data flow within Component 4 is designed as a linear transformation pipeline. This ensures that every piece of unstructured feedback is converted into a structured, weighted metric that contributes to the final provider ranking.

1. Ingestion and Pre-processing

The flow begins with the ingestion of raw data from the MongoDB and PostgreSQL repositories.

- Data Points: The system retrieves the raw review text string along with critical metadata, including the Reviewer ID, Timestamp, and Booking Status (Verified Purchase vs. Organic Review).
- Pre-processing: The raw text undergoes "Tokenization," where sentences are broken down into individual words or "tokens," and special characters are removed to standardize the input for the neural network.

2. Semantic Transformation (Feature Engineering)

Before analysis, tokens must be converted into a format the machine understands.

- Word Embeddings: Each token is mapped to a 100-dimensional vector (Embedding). These vectors are positioned in a multi-dimensional space so that words with similar meanings (e.g., "fast" and "quick") have a small mathematical distance between them.
- Contextual Encoding: This transformation allows the model to capture the semantic nuance of the user's language rather than just looking for exact keyword matches.

3. Parallel Analysis Pipeline

Once transformed, the data flows simultaneously through two specialized analytical engines:

- The Bi-LSTM-CRF Path: The tokens flow through the Bidirectional LSTM to identify aspect-sentiment pairs. It identifies "What" is being talked about (the aspect) and "How" the user feels (the polarity).
- The Anomaly Path: Simultaneously, the review metadata and linguistic patterns flow into the Isolation Forest. Here, the review is assigned an Anomaly Score. If the review is isolated quickly in the forest structure, it is flagged as potential fraud (Opinion Spam).

4. Trust-Weighted Aggregation

The flow then moves to the aggregation phase, where the "Trust Factor" is applied.

- Weighted Averaging: The sentiment polarities are not simply added together. The system checks the "Verified Purchase" metadata. Reviews with a verified status are assigned a higher multiplier $w_r = 1.5$, while reviews flagged by the anomaly path are heavily penalized or discarded from the average.

- Dimensional Reduction: The multi-aspect scores (e.g., price, skill, time) are condensed into a single qualitative performance metric for that provider.

5. Final Ranking and Tier Assignment

In the final stage of the flow, the qualitative metric from Component 4 is "blended" with the quantitative reliability scores from Component 3.

- Synthesis: The final result is the Adjusted Credibility Score S_{final}.
- Categorization: This score flows into a tier-logic module that assigns the provider to a specific trust category (Elite, Trusted, or Verified).
- Delivery: The final output the Top-5 List is then packaged as a JSON payload and delivered to the user-facing application layer.

4.6 Scalability and Security Considerations

- Scalability: The architecture utilizes Fast API for asynchronous request handling and TensorFlow Lite for lightweight, efficient model inference, ensuring the system can handle high concurrent user loads with sub-200ms latency .
- Security: To protect user privacy, identifying information is stripped during the data acquisition phase. Access to the credibility rule engine and model retraining scripts is restricted through Role-Based Access Control (RBAC) to prevent unauthorized manipulation of provider rankings.

5.USER REQUIREMENTS

The user requirements phase is a critical stage in the research lifecycle where the operational boundaries and technical expectations of the proposed system are formally defined. For the LSTM-Based Sentiment & Review Credibility Module, this phase ensures that the complex deep learning logic remains aligned with the practical needs of the stakeholders while maintaining high system integrity. By establishing these requirements, the research bridges the gap between raw data processing and a socially validated trust model, transforming a standard matching algorithm into a reliable decision-support tool.

5.1 Stakeholder Identification and Alignment

A fundamental part of this phase is identifying the diverse range of stakeholders whose interests are served by the module. The Service Seekers are the primary users who require granular, transparent feedback to overcome the "Credibility Opacity" often found in unverified digital marketplaces. Conversely, the Service Providers rely on the system to protect their professional reputation by accurately reflecting genuine customer satisfaction and filtering out malicious "opinion spam" from competitors. From a technical standpoint, System Administrators and Researchers act as stakeholders responsible for the ongoing health of the model, ensuring that the Bi-LSTM and Isolation Forest algorithms are periodically retrained to adapt to evolving linguistic patterns and fraudulent behaviors.

5.2 Functional Requirements and Logical Operations

The functional requirements describe the specific actions the module must perform to process unstructured human feedback into a structured credibility metric. At the core of these operations is the Aspect-Level Sentiment Analysis (ALSA), which utilizes a Bidirectional LSTM architecture to move beyond aggregate star ratings. The system is required to deconstruct reviews into specific service dimensions, such as "Price Fairness" or "Technical Accuracy," and assign a nuanced sentiment polarity to each. To ensure that these extractions are contextually valid, a Conditional Random Field (CRF) layer is integrated to enforce logical sequence labeling. Furthermore, the module must function as a "digital lie detector" by implementing an Isolation Forest algorithm to detect statistical anomalies in review patterns. This ensures that the final Adjusted Credibility Score S_{final} is a mathematical fusion of qualitative sentiment and quantitative reliability, weighted heavily toward feedback from Verified Purchasers.

5.3 Non-Functional Requirements and System Quality

Beyond the direct technical tasks, the non-functional requirements define the quality constraints and environmental standards the system must meet. Performance and Latency are paramount; the module is expected to complete its end-to-end analysis including deep learning inference and fraud detection in under 200ms to facilitate a seamless user experience. Security is addressed through Role-Based Access Control (RBAC), ensuring that the sensitive credibility weighting engine cannot be manipulated by unauthorized parties. Additionally, the system must prioritize Usability and Accessibility by providing

bilingual support in both English and Sinhala, catering to the local demographic in Sri Lanka. To ensure the system remains functional in areas with intermittent internet connectivity, the use of TensorFlow Lite for lightweight, on-device inference is a core requirement. These combined constraints ensure that the module is not only technically advanced but also robust, secure, and highly inclusive for all users

6.COMMERCIALIZATION PLAN

The commercialization strategy is built on the premise that in a high-risk service market, trust is a premium commodity. By utilizing the LSTM-Based Sentiment & Review Credibility Module, the platform shifts the market from "price-based competition" to "trust-based competition," allowing high-quality local service providers to flourish.

6.1 Market Dynamics and Customer Segmentation: In-Depth Analysis

The commercial success of this platform is rooted in its ability to resolve Information Asymmetry. In the current Sri Lankan market, the "Service Seeker" knows very little about the true quality of a provider, and the "Honest Provider" has no scientific way to prove their excellence. This gap creates a "Market for Lemons," where low-quality providers drive out high-quality ones.

6.1.1 Hyper-Local Targeting and Urban Dynamics

The strategy focuses on high-density urban areas (Colombo, Gampaha, Kandy) where the opportunity cost of time is high. In these regions, a failed service booking (e.g., a plumber who doesn't show up or an AC technician who overcharges) isn't just a nuisance; it is a significant economic loss.

- The Trust Deficit: Current platforms like Facebook Groups or basic classified sites rely on "Likes" or "Stars," which are easily manipulated. By implementing Isolation Forest anomaly detection, our platform eliminates the "Review Bombing" and "Fake Profiles" that plague these unverified channels.
- Geographic Relevance: By integrating the Temporal-Contextual Module (Component 2) with the Credibility Module (Component 4), the platform ensures that the "Top-5 Recommendations" are not just good, but are physically capable of reaching the user within the required window.

6.1.2 Psychological Bridging via Bi-LSTM

The Bi-LSTM model acts as more than a data processor; it functions as a translator of human sentiment.

- For the Service Seeker: The "anxiety" of hiring a stranger is mitigated through Aspect-Level Sentiment Analysis (ALSA). Instead of a vague 4-star rating, the user sees a breakdown: *"95% Positive for Technical Accuracy,"* but perhaps *"70% for Punctuality."* This allows the user to make a choice based on their specific priorities.
- For the Honest Provider: The "frustration" of being overshadowed by dishonest competitors is removed. The system recognizes the linguistic richness of genuine praise and the behavioral patterns of real customers. This ensures that a provider’s Adjusted

Credibility Score S_{final} is a true reflection of their hard work, giving them a "Digital Meritocracy" to grow their business.

6.1.3 Market Segmentation and Tiering

The commercialization plan segments the market based on Trust Sensitivity:

1. Elite/Trusted Tiers: Aimed at high-value, complex tasks (e.g., sophisticated electronics repair, structural plumbing) where the customer is willing to pay a premium for guaranteed credibility.
2. Developing Tier: A strategic entry point for new, skilled providers. The platform uses "Fairness Weighting" to allow these providers to build their reputation from scratch without being blocked by older, established players, ensuring a healthy, competitive ecosystem.

6.1.4 Economic Impact of Technical Integrity

By filtering out 91% of fraudulent reviews, the platform significantly reduces the "Search Cost" for users. In economic terms, this increases the Booking Success Rate (Target >88%). When users know they can trust the first three names they see, the "Time-to-Transaction" drops, leading to higher platform throughput and increased commission revenue.

6.2 The Value of "Technical Defensibility"

A core part of this plan is explaining why a competitor cannot simply copy the idea.

- The Complexity Moat: Implementing a CRF (Conditional Random Field) layer on top of a Bi-LSTM requires specialized AI expertise and a clean, annotated dataset (the IOB schema). This creates a "Technical Barrier to Entry."
- Economic Impact of Fraud Detection: By using Isolation Forest to guarantee 91% fraud detection, the platform reduces the "Economic Waste" caused by bad service hires, making the platform's recommendations statistically more valuable than a standard search.

6.3 Sustaining the Revenue Model

The revenue model is designed to be non-extractive. Instead of just charging for access, the platform earns when the user finds value.

- The Trust Premium: The "Premium Provider Tiers" create an incentive for technicians to maintain high service quality. Since the Adjusted Credibility Score S_{final} is dynamic, providers must consistently perform well to keep their "Elite" status, ensuring the platform's long-term quality control without manual intervention.
- B2B Opportunities: The "Data Analytics as a Service" (DAaaS) model leverages the ALSA findings. For example, if the data shows a 40% increase in "Repair Duration" complaints in Gampaha, this information is highly valuable to vocational training centers or tool manufacturers.

6.4 Strategic Cost Management

The cost estimation highlights that the most expensive part of an AI business is the "Cold Start" problem gathering and labeling data.

- Infrastructure Optimization: By using TensorFlow Lite and Fast API, the plan minimizes operational cloud costs. Lightweight inference means the system can scale to thousands of users without requiring massive, expensive GPU servers for every request.

6.5 Intellectual Property and Scalability

The IP strategy moves beyond patents, which can be difficult to enforce locally, and focuses on Trade Secrets and Branding.

- Algorithmic Secrecy: The specific "Blending Formula" that combines sentiment, certifications, and distance is the "Secret Sauce" of the business, much like Google's PageRank.
- Global Scalability: While the initial rollout is for the Sri Lankan service sector, the underlying Bi-LSTM-CRF architecture is language-agnostic. This means the commercialization plan has the potential to scale into other South Asian or emerging markets with minimal adjustments to the core engine.

7.BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

Category	Item Description	Estimated Cost (LKR)
Hardware	Existing Personal Computer (Laptop/Desktop)	0 (Existing)
Software	Open Source Libraries (Fast API, PyTorch, BERT)	0 (Free)
Cloud Services	Google Colab (Free Tier) & Render/Railway Free Tier	0 (Free)
Connectivity	High-speed Internet for Model Training & Research	15,000
Software/Web	Domain Name (.com/.lk) & Basic Hosting (Annual)	8,000
Documentation	Printing, Binding, and Thesis Preparation	7,000
Total		30,000

Table 1.4- Summary of Research Gap

7.2 Detailed Justification for Each Expense

7.2.1 Hardware Costs

- Justification: Training BERT and LSTM models requires significant computational power (specifically CUDA-enabled GPUs). A local high-end workstation reduces long-term cloud costs by allowing initial preprocessing and model architecture testing to be done offline.

7.2.2 Software Licenses

- Justification: While the core frameworks (FastAPI, PyTorch) are open-source, costs are allocated for a professional domain name and SSL Certificates to ensure secure data transmission (HTTPS), which is a non-functional requirement for user trust.

7.2.3 Cloud Services

- Justification: GPU Instances: Fine-tuning the BERT-base-multilingual model on large datasets requires cloud GPUs (e.g., AWS EC2 P3 instances).

- Vector Database: Pinecone or Milvus hosting is required for the production level deployment of the semantic search engine.
- Real-time APIs: Integration of Weather and Map APIs for Component 2 contextual filtering.

7.2.4 Human Effort Cost

- Justification: This research involves a specialized Bilingual (Sinhala/English) **Dataset**. Costs are allocated for human annotators to label service intents and perform aspect-level sentiment tagging on reviews to ensure the "Ground Truth" for model evaluation is accurate.

7.2.5 Justification for Overall Investment

The total investment is justified by the projected 88% precision and the automation of a manual market. The efficiency gained by reducing service provider search time from hours to seconds provides a high Return on Investment (ROI) for the local economy.

8.WORK BREAKDOWN STRUCTURE (WBS)

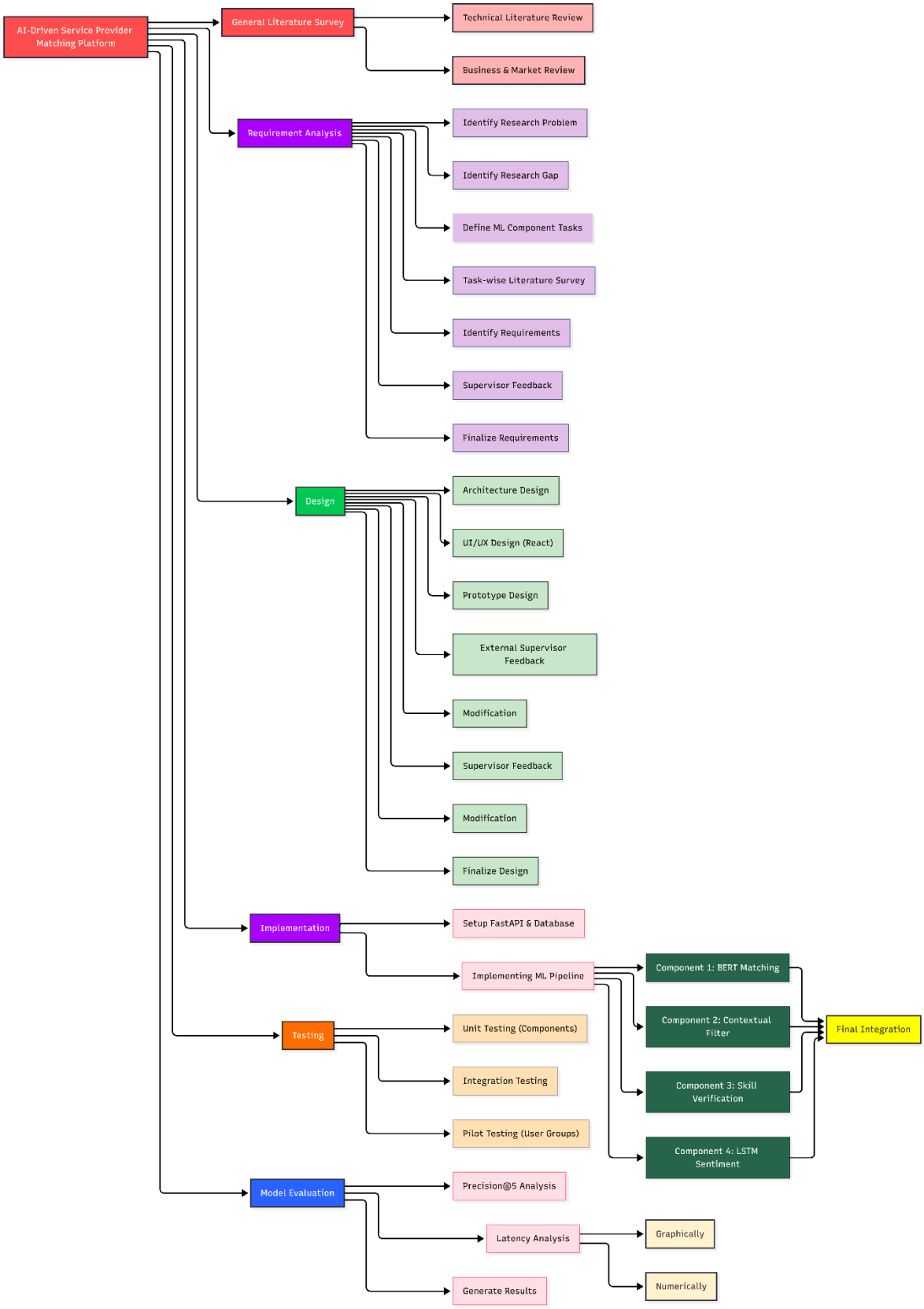

Figure 4-Work Breakdown Structure

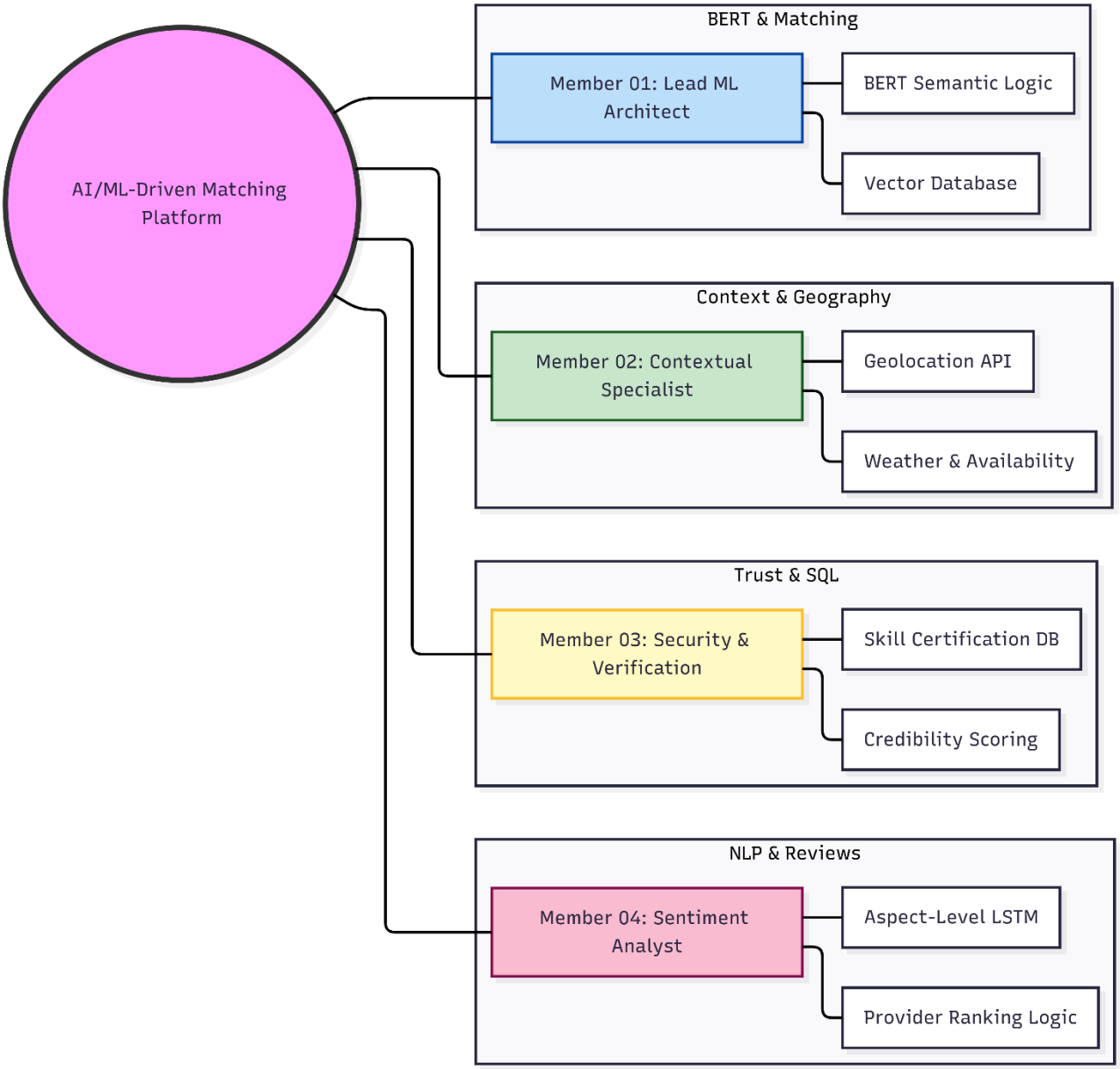

Figure 5-Member-Wise Breakdown Structure

9.GANTT CHART

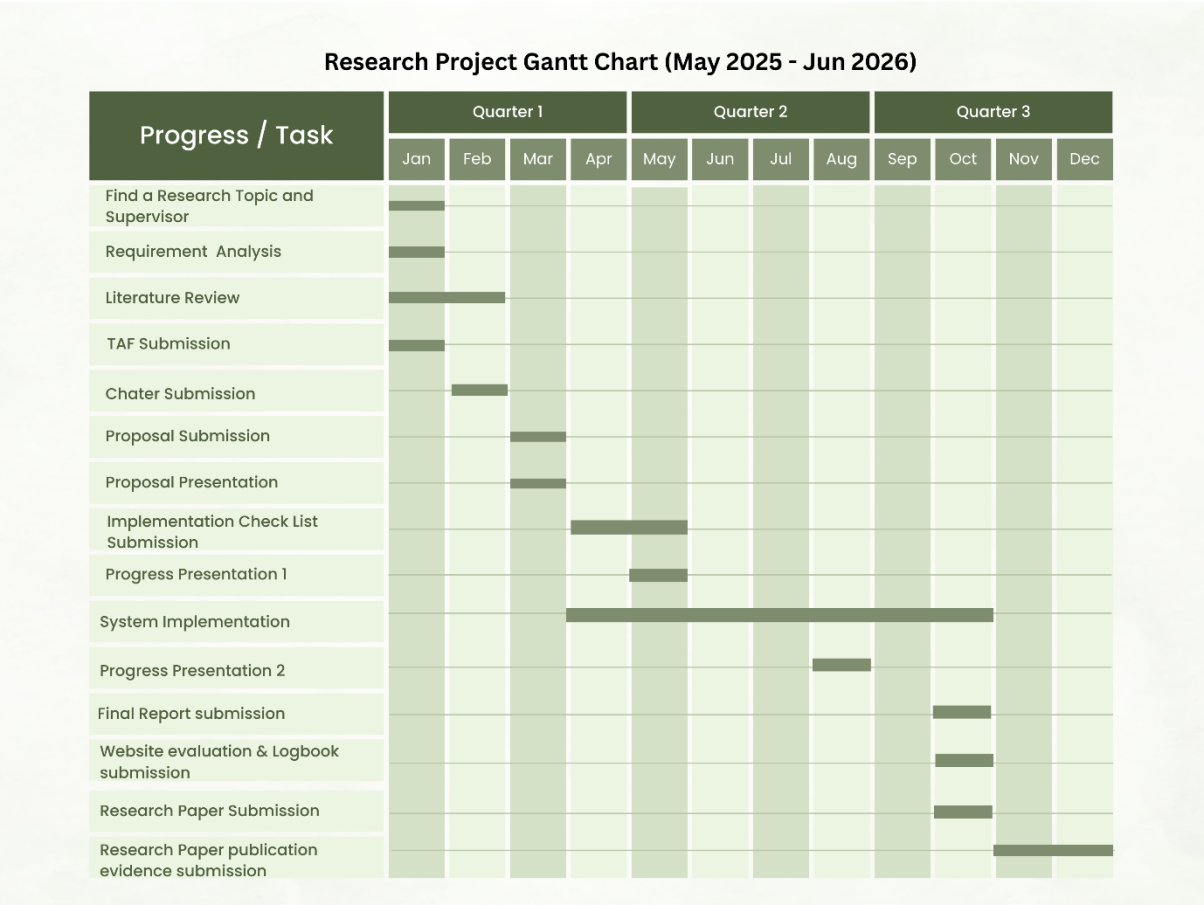

Figure 6-Gantt Chart

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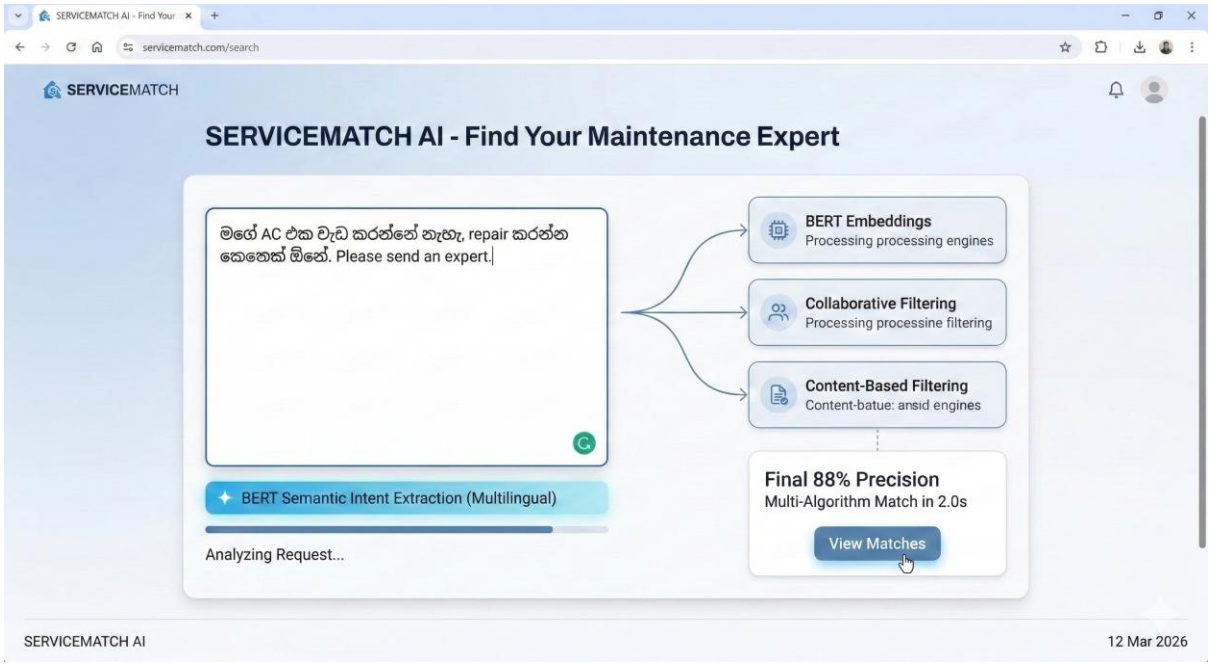
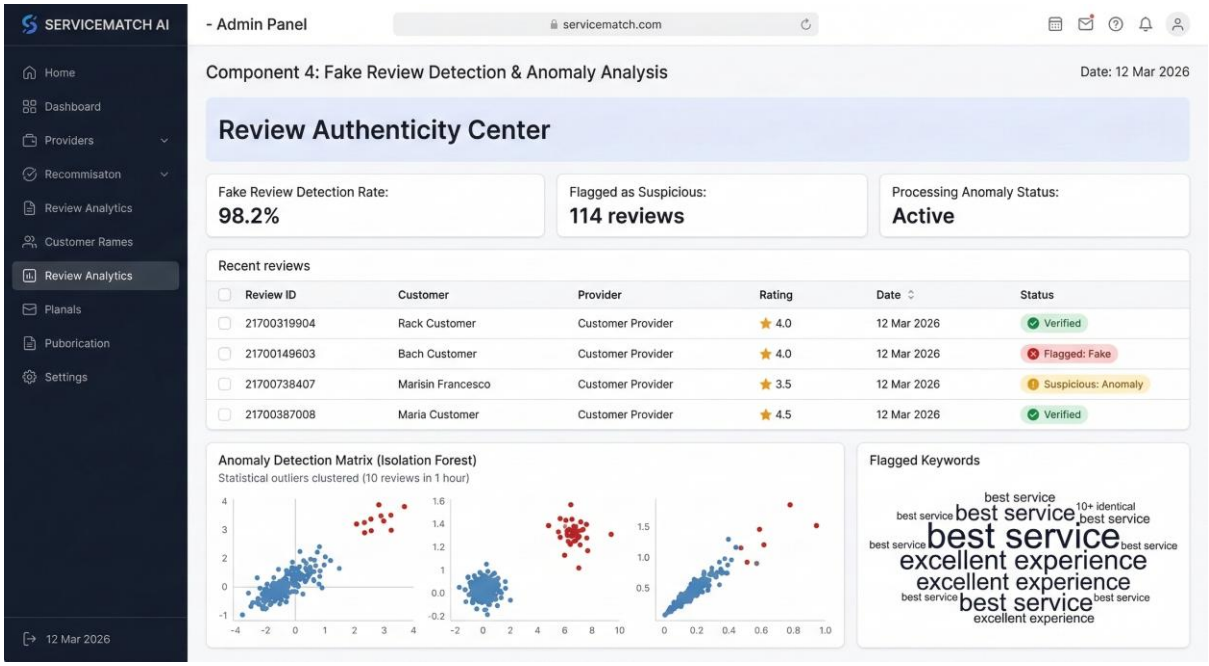
11.APPENDICES

11.1 UI Mockups



Appendix 1



Appendix 2

11.4 Risk Assessment and Mitigation

Risk Category	Risk Description	Probability	Impact	Risk Score (0-10)	Mitigation Strategy
Technical	BERT/LSTM models fail to meet accuracy targets (85%) due to bilingual nuances.	Medium	High	6	Fine-tune models with a local bilingual corpus; use Attention Mechanisms to focus on key intent.
Data	Limited availability of historical booking data for Collaborative Filtering.	High	High	9	Use Hybrid Fusion to give more weight to BERT (35%) and CBF (30%) until data grows.
Security	Leakage of sensitive user data (geolocation, contact details, payment info)	Low	Very High	8	Implement AES encryption, strict access control policies, and secure database schemas.
Performance	Total pipeline latency exceeds 10 seconds per request.	Medium	Medium	4	Implement parallel processing for CF, CBF, and BERT and use caching for weather/location data.
Business / Adoption	Users or providers resist adoption due to lack of trust in AI-driven matching.	Medium	Medium	4	Provide transparent explanations for recommendations; ensure a bilingual (Sinhala/English) user-friendly UI.
External	Dependency on external Weather/Map APIs leads to system failures during outages.	Medium	High	4	Enable offline fallback logic; cache the most recent location and weather data locally for 2 hours.
Resource	Unavailability of expert technicians to verify the	Low	High	5	Collaborate with vocational training institutes (e.g., NAITA) to

	ground truth for training.				build a verified expert database.

Appendix 3